

Chapter 3: Probability and Statistics

The language you need before you can talk about data

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Chapter 1

Probability and Statistics

The language you need before you can talk about data

At this point we're going to start doing data science, and it's worth stopping to ask what data actually *is*. To answer that, we need the language of probability and statistics.

Some of this you will have seen before. That's fine — it's often good to start from the basics even when you know them, because coming back to something a second time is where the deeper understanding happens. This chapter makes no assumptions. It starts from the definition of a probability and ends with everything you need to look at a column of numbers and say something defensible about it.

The chapter is deliberately mathematical and has almost no code in it. The code comes next: the notebook that follows this chapter builds random number generators, histograms, and Monte Carlo integration from scratch, and every algorithm in it is an implementation of something derived here.

1.1 Some terms

I use a lot of these words loosely in conversation, and when I wrote this down I had to take a moment to work out what I actually meant by them. So let's be careful, at least once.

Probability is a mathematical framework for quantifying the likelihood of future events. How likely is it to rain tomorrow? You can answer with a number — 50% — and that number means something.

Though be careful about *what* it means. When the forecast says a 50% chance of rain, that is usually a statement about 50% of the *area* getting rain during that window. It is not a statement that your personal probability of getting rained on is one half. The two are related, and the forecast is still useful to you, but they are different claims. This distinction — between the quantity someone computed and the question you actually wanted answered — is going to come back repeatedly.

Statistics is the analysis of data using the language of probability. Why would you do that? To answer questions, to quantify uncertainty, and to predict future events. An almanac that tells

you the average temperature for a given week is the output of a statistical analysis of a decade of past data, compressed into a single summary number — and that number is good enough to plan around. It isn't the probability of anything by itself, but it tells you the most likely thing to expect, and it tells you that if what happens is far from it, something unusual has occurred.

Data is observations or measurements of past events.

An **experiment** is a controlled setup built to test a specific hypothesis. You arrange things as carefully as you can, take the data, and determine an answer to the question you started with.

Data collection can itself be treated as a kind of experiment. Often you aren't running a controlled setup at all — the data is simply arriving, or it already exists, and you decide later what question to ask of it. Analysing a decade of weather patterns is happening *now*, not back when the thermometers were read. And if the question turns out to require pulling records from several different sources and assembling them, then finding and combining that data *is* the experiment. Once you have it in hand, the two situations look much the same. But it's worth knowing which one you're in.

1.1.1 Models

The word **model** comes up constantly, and it means several different things.

A **theoretical model** captures assumptions or our best understanding, possibly with little or no quantitative input from data. This is what a physicist is doing when they try to reduce everything to the smallest possible set of rules, laws and constants such that everything else follows. A theoretical model may describe things we've never seen. It may be deliberately unrealistic — a limiting case, or a thought experiment — and still be useful. Whether it corresponds to reality is a *separate* question, answered by experiment. A theoretical model is a theoretical model; testing it is a different step.

An **empirical model** is the opposite. Here you don't claim to know anything about the underlying mechanism. You have some technique that takes previous data, characterises it, and predicts future data. It just fits. An empirical model can be excellent or terrible, and again, determining which is a separate process.

Statistical modeling predicts the probability of future events using previous data. There's overlap with empirical modeling — a ball-and-stick model of a molecule is empirical, it isn't what is really happening, but it's right often enough to be worth having — while a statistical model specifically goes after the underlying probabilities, and then uses those probability distributions to say something about the future.

A **generative model** creates new data. A **simulation** uses theory to create new data, and doesn't have to be statistical at all: simulating billiard balls can be completely deterministic, with a single answer. A **Monte Carlo simulation** is one that is statistical — somewhere inside it, it is rolling dice. What that means concretely is that it knows the underlying probability distributions and *samples* them to produce data. You'll write one of these yourself.

Generative AI sits somewhere between the last two: it's something like an empirical Monte Carlo simulation. You take data, learn its probability distributions, and then sample those distributions to make new data. That is a sentence we will come back to and unpack properly

much later in the course, but it is worth planting now, because it is genuinely all this chapter is about.

1.1.2 Science, and why any of this matters

At its core, science is using data to test and improve theoretical models. That's true whether the field is highly quantitative or much more qualitative.

The process runs like this. Propose a model. Devise an experiment to test it — and when you do, aim at where you think the model is *weak*. If you test somewhere you already know it works, you learn nothing. Make a prediction: if my model is right, this experiment gives that outcome. Run the experiment. Perform a statistical analysis of what the data actually says. Your measurement comes back as 5 ± 1 ; your theory said 4.5; now you do a test to ask whether those are consistent. Then you either learn something new or you learn what the next experiment should be, and you go around again.

One thing worth saying plainly: **most of science is eliminating bad hypotheses**, and that is how the field actually advances. A large particle physics collaboration might publish on the order of two hundred papers a year and discover one major new particle in a decade. The value of the rest is that an enormous space of possible theories has been tested and ruled out. Knowing where something *isn't* tells you where to look next. If every experiment succeeded, we wouldn't need many scientists.

Outside science the same machinery applies: use previous data and models to predict future events. That, in the end, is why anyone wants to hire a data scientist.

1.2 Probability theory

Probability theory analyses the frequency of **events** — of something happening. “Event” is a generic term; in particle physics every collision is called an event, and elsewhere you might say “a sample,” but the idea is the same.

The probability p of an event x happening means: if I make repeated observations, p is the fraction of the time that x is the outcome. The probability of heads on a coin is 50%, meaning that if I flip it many times, half the time I get heads.

Take another example: the frequency of a disease in a population is 1 in 1000. Interpreted this way, it means that if you go out and pick a person at random, one time in a thousand that person has the disease. This is the **frequentist** reading of a probability.

But now suppose you take a test for that disease and it comes back positive. Do you have it? You aren't going to repeat the test a thousand times — and if there's something systematically wrong, repeating it might give the same wrong answer anyway. You have one result. How do you interpret it?

You can still use probability here. Combining the accuracy of the test with the frequency of the disease in the population gives you a **degree of belief** in the result. The test says positive, but

taking everything into account the chance you're actually sick might be far lower than you'd guess. This is the **Bayesian** reading.

These are two different ways of interpreting what a probability *is*. They are not two different mathematics. The equations are the same and the computations are the same; they give the same answers. What differs is what you take the number to mean.

1.2.1 Where uncertainty comes from

Before the machinery, it's worth naming what we're quantifying. An analysis has three jobs: estimate a parameter, quantify the uncertainty on that estimate, and test the extent to which a prediction agrees with the data. The second is the one people skip, and it's the one that makes the other two mean anything.

Uncertainty has three sources:

1. **Stochastic** — genuine underlying randomness. Quantum mechanics is the pure case; the shuffle of a deck of cards is the everyday one.
2. **Incomplete observations** — you can't see everything. The Monty Hall problem is the classic puzzle here: the answer depends entirely on what the host knows and you don't.
3. **Unknowns** — incomplete modeling. You don't know what you left out.

There is a fourth that doesn't appear on that list and is usually the one you actually spend your time on: the **precision of your own apparatus**. Drop a ball and time its fall, and your measurements will scatter — not because of anything quantum, but because the drop height isn't exactly one metre every time, and because your thumb on the stopwatch is inconsistent. Start and stop the clock with a laser instead and the spread narrows.

That distinction matters. Stochastic uncertainty is in the thing you are measuring and no instrument will remove it. Apparatus uncertainty is in *you*, and better equipment or more care will shrink it. The first you can only characterise; the second you can also attack.

Both, though, are handled the same way — and this is the idea the rest of the chapter is built on. Repeat the measurement many times and histogram the results. That distribution is telling you about your experiment: how good the timing was, how consistent the release was. Its centre is your best estimate and its width is your uncertainty. Which is the real reason you repeat an experiment: not to be more sure of one number, but to *measure how unsure you should be*.

We'll come back to exactly this at the end of the chapter and do it properly.

1.3 Basic definitions

This part is abstract, and it may not be obvious what it's for until the examples arrive. Bear with it.

Start with a set S and subsets A and B inside it. Think of S as a region marked out on a table, representing the space of everything that could possibly happen. If you stand above the table and drop a coin, S is “the coin lands somewhere on the table,” and A and B are sub-regions of

it. If everything is symmetric, the probability of landing in a region is just proportional to its area — bigger region, more likely. Area isn't probability in general, but in this picture it is, and that's the analogy the drawing is carrying.

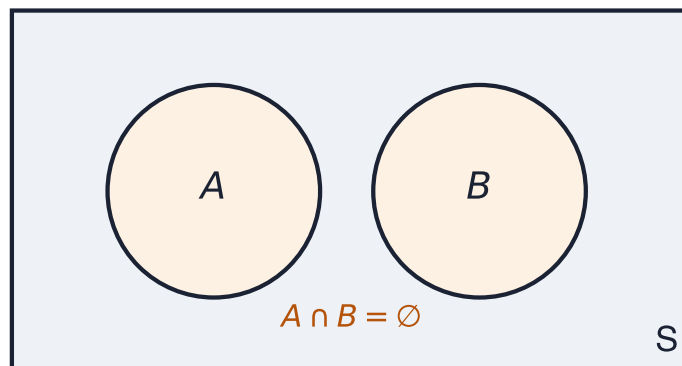


Figure 1.1: Two disjoint subsets of the sample space. S is everything that can happen; A and B are regions inside it that do not overlap.

Now the rules.

Something must happen, so the probability of the whole space is one:

$$P(S) = 1$$

For any subset $A \subset S$, the probability is non-negative — and at most one, reaching one only if A is all of S :

$$0 \leq P(A) \leq 1$$

The bar means “not.” The probability of *not* being in A is whatever area is left over:

$$P(\bar{A}) = 1 - P(A) \quad P(A \cup \bar{A}) = 1$$

Getting nothing at all has probability zero:

$$P(\emptyset) = 0$$

If A and B don't overlap — if $A \cap B = \emptyset$ — then the probability of A *or* B is just the sum:

$$A \cap B = \emptyset \Rightarrow P(A \cup B) = P(A) + P(B)$$

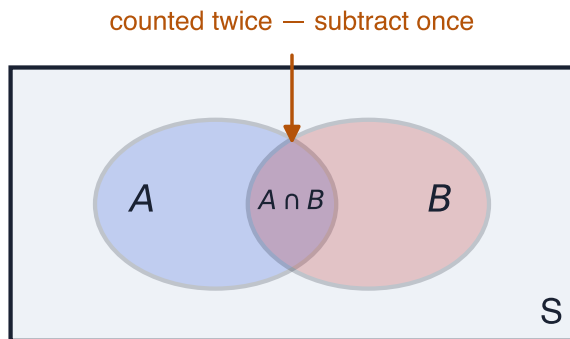
If A sits entirely inside B , then A can't be more likely than B :

$$A \subset B \Rightarrow P(A) \leq P(B)$$

And in the general case, where they overlap:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

You add the two areas and then subtract the overlap once, because otherwise you have counted it twice.



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Figure 1.2: Overlapping subsets. Adding the two areas counts the lens where they cross twice, which is why $P(A \cap B)$ is subtracted once.

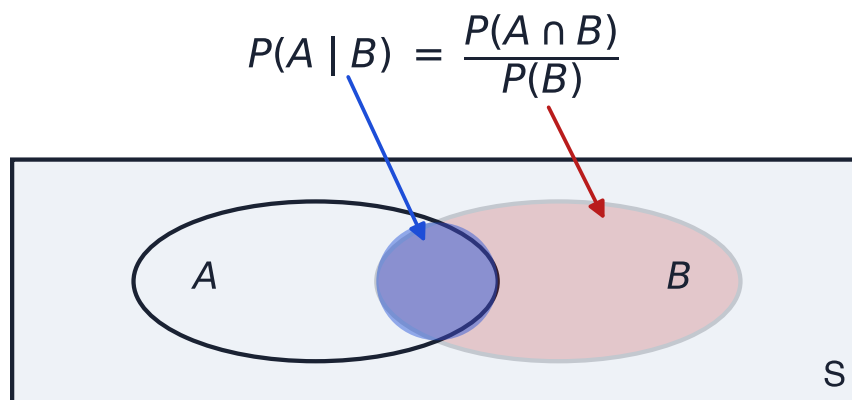
Don't get hung up on the symbols. $\cup$ is "or," $\cap$ is "and," and the picture is what you should be carrying around.

1.4 Conditional probability

What is the probability of A *given* that I already know B is true?

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Look at the picture to see why. I've told you we are inside B . That's now certain — B has probability one. So I have to renormalise everything relative to B , which is what dividing by $P(B)$ does. Within that restricted world, the part that is also A is $A \cap B$. Asking for the probability of A given B is a different question from asking for the probability of A within S , and the division is exactly the correction between them.



conditioning on B throws away everything outside B ,
then asks what fraction of what is left is also A

Figure 1.3: Conditioning on B renormalises the sample space to B . The numerator is the part of B that is also A .

1.4.1 An example

I roll a six-sided die. I don't tell you what I got, but I tell you it was three or less. What is the probability that I rolled a one?

The space of possibilities S is $\{1, 2, 3, 4, 5, 6\}$. The event B — what I've told you — is $\{1, 2, 3\}$. The event A — what you want — is $\{1\}$.

$$P(A | B) = \frac{P(1 \text{ and } \leq 3)}{P(\leq 3)} = \frac{1/6}{3/6} = \frac{1}{3}$$

The numerator is $1/6$: rolling a one *and* rolling three or less is just rolling a one, since a one is already three or less. The denominator is $3/6 = 1/2$: three of the six faces qualify. The answer is one in three, which is what you'd expect — you've been told the roll is one of three equally likely faces.

1.5 Independence

Two events are **independent** when knowing one tells you nothing about the other. The test is whether the probabilities factorise:

$$P(A, B) = P(A) P(B)$$

where $P(A, B)$ means “ A and B both happen.” If that holds, then the conditional probability collapses:

$$P(A | B) = \frac{P(A)P(B)}{P(B)} = P(A)$$

Knowing B has changed nothing about A , which is exactly what independence should mean.

A correction worth making explicitly. It is tempting — and I have said this in lecture — to reach for “ A and B don’t overlap” as the condition for independence. It isn’t. Disjointness is close to the *opposite* of independence: if A and B can’t both happen, then learning that B occurred tells you A definitely did not, which is about as far from “tells you nothing” as you can get. Formally, if $A \cap B = \emptyset$ then $P(A \cap B) = 0$, so $P(A | B) = 0$, not $P(A)$. Independence is the factorisation above and nothing else. We’ll see it stated correctly again for continuous variables later in this chapter.

1.5.1 The two interpretations, again

With conditional probability in hand, the frequentist and Bayesian readings can be stated precisely.

Relative frequency. $A, B, \dots$ are outcomes of a repeatable experiment, and

$$P(A) = \lim_{n \rightarrow \infty} \frac{\text{number of times the outcome is } A}{n}$$

This is the best way to assess a probability *from data*: run it many times and count.

Subjective probability. $A, B, \dots$ are hypotheses — statements that are true or false. $P(A)$ is your degree of belief that A is true. “Do I have COVID?” is not a repeatable experiment. You took one test, it said something, and $P(A)$ is how much you believe it.

1.5.2 The two readings are two conditional probabilities

Here is the sharpest way to see the difference, and it is worth sitting with, because it is what the rest of this chapter turns on.

Take one medical test. It has two conditional probabilities attached to it, and they are not the same number:

$$P(+ | \text{sick}) \quad \text{and} \quad P(\text{sick} | +)$$

The first is the test’s **true positive rate**. It is a frequentist quantity through and through: take a large number of people who are known to be sick, test them all, and count the fraction that come back positive. That is a repeatable experiment, the limit is well defined, and it is a property of the *test*. It is the number a manufacturer measures and puts on the box.

The second is the one you care about when your own result arrives. And it is not a frequency of anything, because the experiment isn’t repeatable — there is one of you, you are either sick

or you are not, and running the test again mostly re-measures the same test. “I am sick” is a **hypothesis**, and $P(\text{sick} \mid +)$ is a degree of belief in it. That is the Bayesian reading.

So the two interpretations are not two rival philosophies applied to the same quantity. They attach to two *different* quantities, and conditional probability is what tells them apart: **which side of the bar is the hypothesis, and which side is the data**.

- Condition on the hypothesis, ask about the data $\rightarrow P(\text{data} \mid \text{hypothesis}) \rightarrow$ a frequency, measurable by repetition
- Condition on the data, ask about the hypothesis $\rightarrow P(\text{hypothesis} \mid \text{data}) \rightarrow$ a belief, and the thing you actually wanted

Which raises the obvious question: you can measure the first one and you want the second one, so how do you get from one to the other? That is precisely what Bayes' theorem does, and it is why the next section exists.

One consequence worth noticing in advance. The conversion is going to need the prior $P(\text{sick})$ — the frequency of the disease in the population, which is a frequentist quantity, measured by counting. So the Bayesian calculation *consumes* frequentist numbers. The two readings aren't in competition; the machinery is shared, the equations are identical, and they give the same answers. What differs is which question you are asking.

1.6 Bayes' theorem

Sometimes you know $P(B \mid A)$ and you want $P(A \mid B)$. This happens constantly, and it is called an **inverse problem**: a test told you the probability of getting a positive result given that you're sick, but what you actually want to know is the probability that you're sick given a positive result.

The derivation is short. Write the conditional probability both ways round:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \quad P(B \mid A) = \frac{P(B \cap A)}{P(A)}$$

The intersection doesn't care about order — $P(A \cap B) = P(B \cap A)$ — so both left-hand sides multiplied out give the same thing:

$$P(A \mid B) P(B) = P(B \mid A) P(A)$$

and therefore

$$P(A \mid B) = \frac{P(B \mid A) P(A)}{P(B)}$$

That's it. It is a way of converting one conditional probability into the other.

1.6.1 The denominator, and the partition

In practice $P(B)$ is the awkward term, and there's a standard way to get it. Suppose the A_i are a set of possibilities that between them cover everything and don't overlap — a **partition** of the sample space:

$$\bigcup_i A_i = S$$

Then B can be chopped up along the same lines:

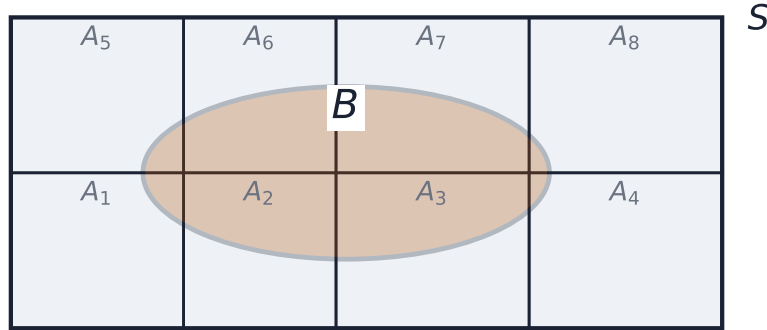
$$B = B \cap S = B \cap \left(\bigcup_i A_i \right) = \bigcup_i (B \cap A_i)$$

and since those pieces don't overlap, their probabilities add:

$$P(B) = \sum_i P(B \mid A_i) P(A_i)$$

This is the **law of total probability**: sum over every distinct way B could come about. Substituting it back gives the form of Bayes' theorem you'll actually use:

$$P(A \mid B) = \frac{P(B \mid A) P(A)}{\sum_i P(B \mid A_i) P(A_i)}$$



$$P(B) = \sum_i P(B \mid A_i) P(A_i)$$

every distinct way B could come about

Figure 1.4: A partition of S into regions A_i , with B laid across several of them. The probability of B is the sum of its pieces.

The two terms have names. $P(A)$ is the **prior** — what you knew before the measurement. $P(A \mid B)$ is the **posterior** — what you know after it. Bayes' theorem is the rule for updating one into the other when data arrives.

1.7 The disease test

This is the example the whole first half of the chapter has been building toward.

There is a disease. From population statistics we know:

$$P(\text{sick}) = 0.001 \quad P(\text{not sick}) = 0.999$$

That's the prior — prior knowledge about the population, before anyone takes a test. One person in a thousand has this.

A test exists, and its performance is characterised by four numbers:

	Test says positive	Test says negative
Sick	$P(+ \text{sick}) = 0.98$ — true positive	$P(- \text{sick}) = 0.02$ — false negative
Not sick	$P(+ \text{not sick}) = 0.03$ — false positive	$P(- \text{not sick}) = 0.97$ — true negative

The diagonal is the test getting it right; the off-diagonal is the two distinct ways of getting it wrong. By any casual reading this looks like a good test. It catches 98% of sick people and only falsely alarms 3% of healthy ones.

You take the test. It comes back positive. What is the probability that you're sick?

The partition here is just $A_i = \{\text{sick}, \text{not sick}\}$ — two possibilities, covering everything. So:

$$\begin{aligned}
 P(\text{sick} | +) &= \frac{P(+ | \text{sick}) P(\text{sick})}{P(+ | \text{sick}) P(\text{sick}) + P(+ | \text{not sick}) P(\text{not sick})} \\
 &= \frac{0.98 \times 0.001}{0.98 \times 0.001 + 0.03 \times 0.999} = \frac{0.00098}{0.03095} = 0.032
 \end{aligned}$$

About 3%. A test that is 98% accurate came back positive, and you are still only three percent likely to be sick.

The reason is in the denominator. There are so many more healthy people than sick ones that the small false-positive rate applied to the huge healthy population produces far more positives than the high true-positive rate applied to the tiny sick population. Ninety-eight true positives against nearly three thousand false ones.

This is the single most useful thing in this chapter, and the reason it is worth learning Bayes' theorem properly. If you test at random for a rare condition, get a positive, and have no symptoms, the most likely explanation is that the test was wrong.

Note the “at random” and “no symptoms.” If you have symptoms, none of these numbers apply: $P(\text{sick})$ is no longer one in a thousand, and the whole calculation shifts. The prior is doing the

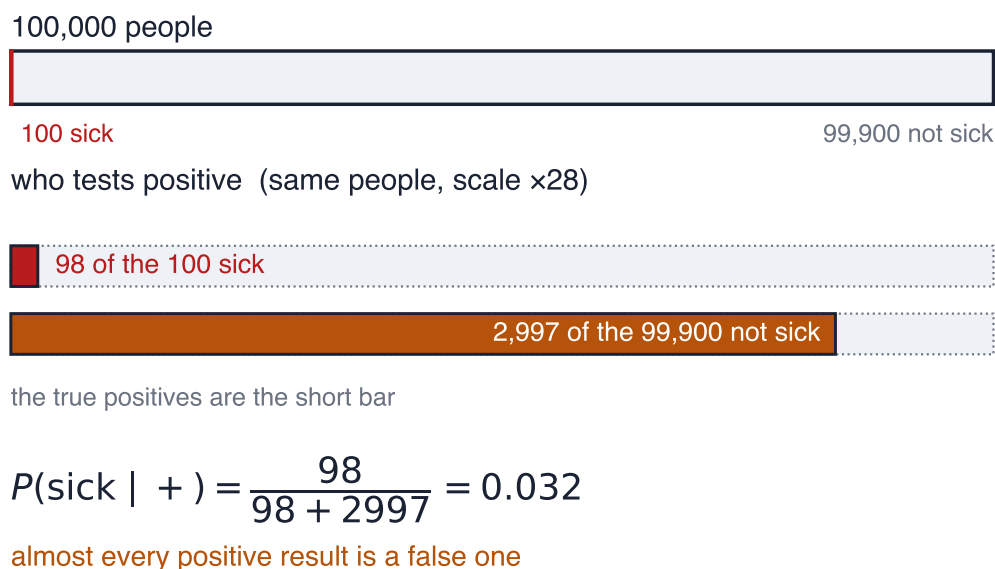


Figure 1.5: Why the answer is 3%. Out of 100,000 people, 98 sick people test positive — and 2,997 healthy people also test positive. Almost every positive result is a false one.

work here, and changing the prior changes the answer completely. That is not a flaw in the method — it is the method telling you, correctly, that context matters.

1.8 Data

Everything so far has been about probability. Now let’s connect it to data.

Suppose I build a table of everyone in this class: name, age, major, GPA, and whatever else.

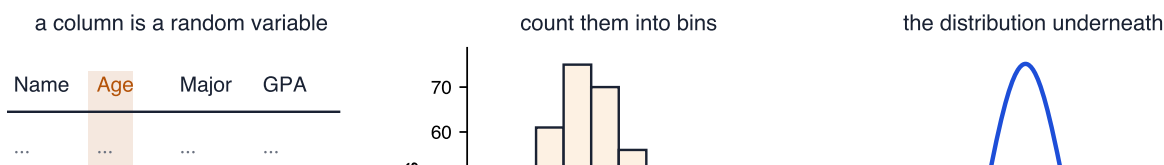
Instance	Name	Age	Major	GPA	...
1	...	...	...	...	
2	...	...	...	...	
3	...	...	...	...	
...					

This is data, and it can be read two ways.

Read **by row**: each row is an **instance**, a single data point — one person.

Read **by column**: each column is a **feature**, some variable measured about each person.

The column reading is the important one. Look at the age column. When I talk about “the age of my students,” I am not talking about any one person’s age — I’m talking about the ages that a student in this room can have, and how often each occurs. That is a **random variable**. To say anything about it, I need to understand the underlying probability distribution: how likely the variable is to take each possible value.



1.9 Random variables

If x is a random variable, I write

$$x \sim P(x)$$

read as “ x is **drawn from** $P(x)$.” The values it can take form a set:

$$x \sim \{x_1, x_2, \dots, x_n\}$$

Every time you get a value, it is as though all the possible numbers are in a bag and you have pulled one out.

Random variables come in two kinds, and the distinction matters more than it first appears.

Discrete. The variable takes one of a countable set of values — a die roll is 1 through 6. Then $P(x)$ is a **probability mass distribution**, and $P(x_1)$ is directly the probability of getting the value x_1 . If the distribution is normalised, you can read the answer straight off the height of the bar.

Continuous. Here something changes. If x can be any real number in a range, then the probability of getting *exactly* x_1 is zero. Asking for it isn’t a meaningful question. What you can ask is the probability of landing in a small interval:

$$P(x \in [x_1, x_1 + \delta x]) = p(x) \delta x$$

and in the limit $\delta x \rightarrow 0$ this defines $p(x)$, the **probability density function**.

Notice the two different letters. Capital P is a probability — a number between zero and one. Lower-case p is a *density*, and it does not give you an answer on its own: you have to multiply it by a range. That’s the whole difference. A density can even exceed one; a probability cannot.

Both readings of a probability carry over to a density, and it is worth writing them out, because this is the form you will actually meet them in:

- **Frequentist.** $p(x) dx$ is the *frequency* with which a measurement of x lands between x and $x + dx$. Repeat the measurement many times, count how often it falls in that window.
- **Bayesian.** $p(x) dx$ is, given the measurement you have, how likely it is that the true x lies between x and $x + dx$.

The same split as before, now attached to the thing you will spend the rest of the course plotting. And the same reassurance: the density is one function, the arithmetic is identical, and only the question changes.

There is a further notational point that trips people up. $P(x)$ with a variable in it means the whole distribution — a function. $P(x = x_1)$, often abbreviated $P(x_1)$, means a single number: the probability of that specific value. $P(x_1) = 1$ means x is always x_1 ; $P(x_1) = 0$ means it is never x_1 .

1.9.1 Properties of a distribution

Whatever P is, three things must hold.

The **domain** of P must be all possible values of x . You have to know what you're computing over, and P needs a value everywhere you might look.

Every probability is between zero and one:

$$\forall x_i \in x : \quad 0 \leq P(x_i) \leq 1$$

And the whole thing is **normalised** — the probabilities of all outcomes sum to one:

$$\sum_{x_i \in x} P(x_i) = 1 \quad (\text{discrete}) \quad \int_{-\infty}^{\infty} p(x) dx = 1 \quad (\text{continuous})$$

1.9.2 The cumulative distribution

Often what you want is not the probability of a value but the probability of being *below* one:

$$F(x_1) = P(x \leq x_1) = \int_{-\infty}^{x_1} p(x') dx'$$

You integrate from the far left up to x_1 . Since p is already normalised, there's nothing to divide by. This is the **cumulative distribution function**.

Its shape follows from the picture. Far to the left there's no area yet, so $F = 0$. Moving right through the bulk of the distribution you accumulate area quickly, so F rises steeply. Past the bulk there's little left to add, so it flattens out and asymptotically approaches one. Every cumulative distribution is that same S-shape, stretched differently.

It's worth learning properly: cumulative distributions turn up again when we evaluate classifiers, and the notebook for this chapter uses one to invert a distribution and generate random numbers from it.

1.9.3 Reading a percentile off the cumulative

Run the definition backwards and the cumulative answers a different, very practical question. Instead of “what fraction lies below x_1 ?”, ask “**below what value does 90% of the distribution lie?**” That value is the 90th percentile, written x_{90} , and it is just

$$F(x_{90}) = 0.9 \quad \Rightarrow \quad x_{90} = F^{-1}(0.9)$$

You almost never have F as a formula, though. You have data. The recipe is the histogram argument again, one step further:

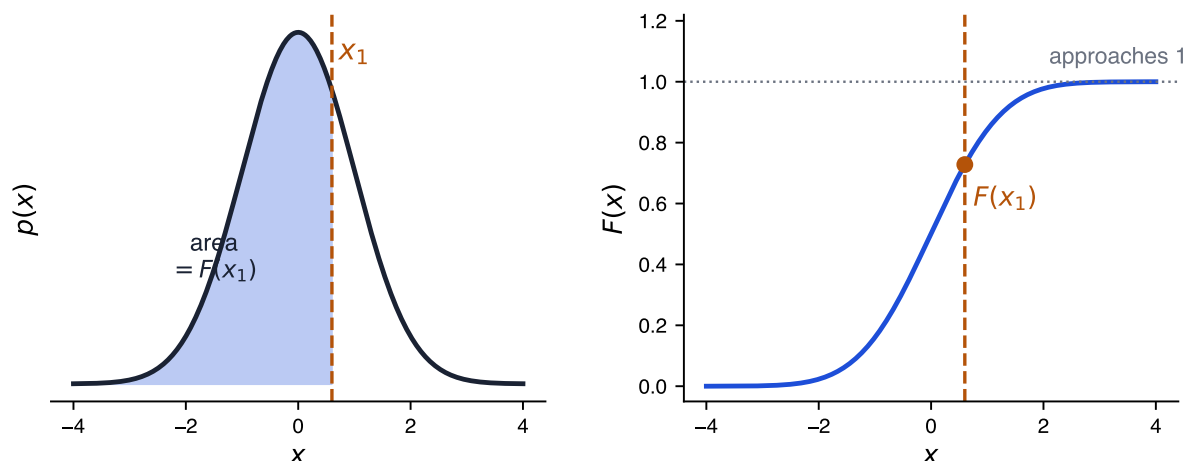


Figure 1.7: A density and its cumulative distribution. $F(x_1)$ is the shaded area to the left of x_1 ; it starts at zero, rises through the bulk, and approaches one.

1. Histogram your sample — this is $n(x)$, a list of counts.
2. Take the **running total** down the list: the cumulative sum. Bin i 's entry is now the number of data points at or below bin i .
3. The last entry is N , the total. Find where the running total first reaches $0.9N$.
4. That bin's edge is x_{90} .

With 1000 points you are looking for where the running total passes 900. Every percentile, median, and quartile you have ever seen quoted is this calculation.

Two things to notice. The answer is only ever a bin edge, so its resolution is your bin width — percentiles from a coarse histogram are coarse. And nothing here assumed a distribution: this works on any sample at all, which is why percentiles get quoted for data that is nothing like normal.

1.10 Histograms

How do you find out what a probability distribution actually is, from data? The answer is a **histogram**, and it is not more complicated than it sounds.

Ask how many people in the room are 18. Count the hands. Ask about 19, count again, and so on. Plot the counts. That is a histogram, and once you have it you can answer questions: if I grab a student at random, how old are they likely to be? The histogram has all of the information.

But a histogram of counts is not yet a probability. If twelve people are 20 years old, the bar reads 12, and 12 is not a probability. To turn counts into probabilities you divide by the total number of people, giving a fraction.

There's one more subtlety: the bin width. If you make the bins twice as wide, each bar catches roughly twice as many entries, and you don't want the answer to depend on a choice you made

about plotting. So you divide by the bin width too. Writing $n(x)$ for the count in the bin at x , N for the total number of data points, and Δx for the bin width:

$$P(x) \approx \frac{n(x)}{N \Delta x}$$

and in the limit of infinite data and infinitely fine bins this stops being an approximation:

$$P(x) = \lim_{\substack{N \rightarrow \infty \\ \Delta x \rightarrow 0}} \frac{n(x)}{N \Delta x}$$

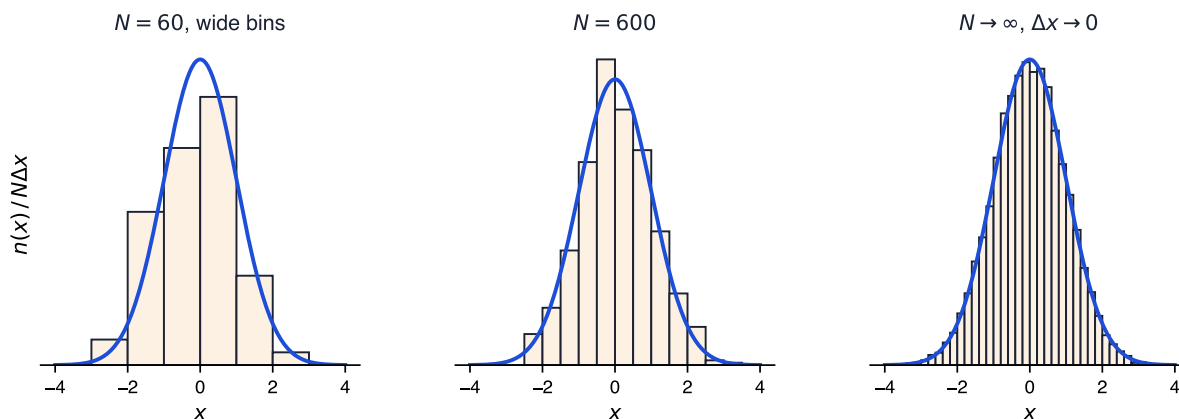


Figure 1.8: Counting into bins. As the sample grows and the bins narrow, the histogram converges on the underlying density.

That formula is the bridge between the two halves of this chapter. On the left is a probability distribution, which is a mathematical object you do algebra with. On the right is a histogram, which is something you compute from a list of numbers. This equation says they are the same thing, and it is the reason a histogram is the first thing you should make when handed a dataset.

And that is genuinely the claim: **if you know the probability distribution underlying your data, you know everything there is to know.** There is nothing more to extract. Every prediction you could make is available from it. So for a single variable, a histogram is very nearly the best you can do.

Only *very nearly*, for one reason we'll come to shortly: sometimes you know in advance what functional form the distribution should take, and then you can do better than a histogram by fitting that form and estimating its parameters. But absent that knowledge, the histogram is it.

1.11 Two dimensions

One variable at a time is not usually enough, because features are related to each other.

Take height and weight. You can histogram height, and you can histogram weight, and each tells you something. But height and weight are correlated — not perfectly, there are short heavy

people and tall light ones, but clearly related — and neither one-dimensional histogram can express that.

Plot one against the other, a dot per person, and you get a **scatter plot**. Divide the plane into bins and count the dots in each, and you get a **two-dimensional histogram**. Everything from the one-dimensional case carries over: divide by the total and by the bin area, and you have a two-dimensional probability density.

Displaying one takes a little thought. You can print the numbers in the cells, which is accurate and hard to read. You can draw a 3D surface. Most often people use a **heat map**, colouring each cell by its count, or a **contour plot**, drawing lines of equal probability. A contour plot is worth getting comfortable with: read it as a topographical map, where the contours are the directions you can move without going up or down. The innermost contour might enclose 68% of the data, the next one more, and so on. Contour plots come back when we look at how models are trained.

1.11.1 Why this matters: the average that isn't

Here is the mistake that this section exists to prevent.

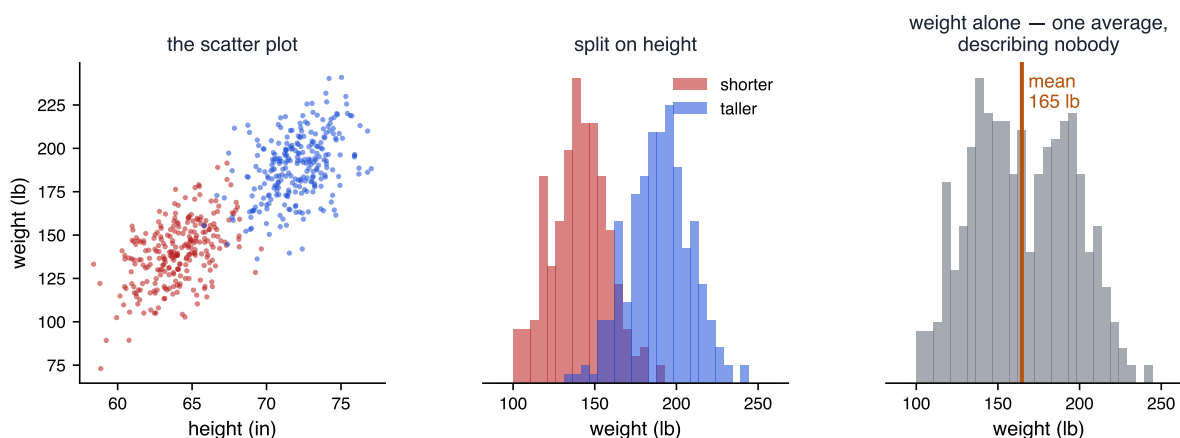


Figure 1.9: Height against weight. Slice the scatter on height and the two weight distributions separate cleanly. Look at weight alone and they merge into one broad shape whose average describes nobody.

Suppose you histogram weight and get a broad, slightly lumpy distribution. You compute the average and report it. But what's actually going on is that there are two populations in there — taller people and shorter people — each with its own weight distribution, and what you're looking at is the two of them superimposed. The average of the combined shape is a number that describes neither group. It may describe nobody at all.

The combined distribution often *shows* you this, if you look. A plateau, a double bump, a shape with shoulders — these are signs that you have more than one population and are averaging across them.

So: **the first thing to do with a new dataset is make histograms of every feature.**

And if there is a categorisation — heart disease versus no heart disease, say — histogram each feature *separately for each category* and overlay them. If the two overlays sit on top of each other, that feature tells you nothing. If they separate, it does. You can look at a new dataset and, in a handful of commands, form a defensible opinion about whether there’s anything in it. That is a real skill and it is almost entirely this one habit.

Think about what you’re really doing when you overlay them: you are looking at $P(x \mid \text{disease})$ and $P(x \mid \text{no disease})$. If you knew those two distributions exactly, you would know everything the data can tell you about distinguishing the two cases.

1.11.2 Multivariate distributions, formally

The one-dimensional definitions generalise directly. For a joint distribution $P(x_1, x_2)$:

Normalisation is a multidimensional integral:

$$\int P(x_1, \dots, x_n) dx_1 \cdots dx_n = 1$$

The **marginal probability** of one variable is obtained by integrating over all the others:

$$p(x_1) = \int P(x_1, \dots, x_n) dx_2 \cdots dx_n$$

This is exactly the projection described above. Integrating the height–weight distribution over weight gives you the height distribution; integrating over height gives the weight distribution. You are summing up everything in the other direction.

And **independence**, stated properly for distributions:

$$P(x_1, x_2) = P_1(x_1) P_2(x_2) \Rightarrow x_1 \text{ and } x_2 \text{ are independent}$$

If the joint distribution factorises into separate distributions for each variable, the variables are independent. One does not depend on the other.

1.12 The curse of dimensionality

Two dimensions worked. Three, four, five — why not keep going?

Because the space empties out. As you add dimensions, the volume between your data points grows enormously. Consider an eighteen-dimensional histogram: the probability of any particular cell containing a data point is very low, and there are an astronomical number of cells. With a finite dataset, almost every bin is empty.

That’s fatal, because an empty bin reads as probability zero, and the probability isn’t zero — it’s just small, and you don’t have enough data to see it. In one or two dimensions you can get

many points into every bin and the histogram is a real measurement. In eighteen you are lucky to have one point in any bin at all, so the histogram tells you almost nothing.

You would need vastly more data. But you can't always get vastly more data — and the amount you need grows exponentially with the number of features.

This is the **curse of dimensionality**, and it is the reason we can't simply make histograms and be done. It's also the reason machine learning exists. Real data is high-dimensional, the thing you want is the probability distribution over all those dimensions, and a histogram cannot give it to you.

Two related consequences, worth knowing the names of. **Distance metrics degrade**: in high dimensions the distances between points become more and more similar to one another, which hurts any algorithm that relies on nearness, such as k -nearest neighbours. And with many features relative to the number of examples, the **risk of overfitting** rises sharply.

1.13 The best you can do

There's a result worth stating now, even though we won't use it for weeks, because it tells you what all of this is *for*.

Suppose you have two hypotheses — H_0 and H_1 , sick and not sick, signal and background — and a measurement $\vec{x}$, which may be a whole vector of features. The **Neyman–Pearson lemma** says that the most powerful test for choosing between them is the **likelihood ratio**:

$$\Lambda(\vec{x}) = \frac{P(\vec{x} \mid H_1)}{P(\vec{x} \mid H_0)}$$

Call it H_1 when the ratio is above some threshold, H_0 when it is below. The threshold sets how cautious you are; the ratio itself is the **test statistic**.

Concretely: a patient arrives, you run every test, and all the results together are $\vec{x}$. Does this patient have the disease? Compute the probability of seeing these results given that they're sick, divide by the probability of seeing them given that they're not, and compare to a threshold. Do that for every patient and you have separated the population as well as it can be separated.

That word — *proven* — is the point. This isn't a good heuristic. For any classification task, if you know the probability distributions, **it is impossible to do better than this**.

Which brings the chapter's argument to its conclusion. The best possible classifier is built from the probability distributions. In one dimension we can measure those with a histogram. In two we can. Beyond that, the curse of dimensionality stops us — and characterising multi-dimensional probability distributions is, precisely, the hard problem in data science. Everything else in this course is a response to it.

1.14 Measurement

A probability distribution is a complicated object: a whole function, with a value everywhere. It would be useful to summarise it in a few numbers.

Start by defining the **expectation value**. Given a random variable x drawn from $P(x)$, and some function f of it, the expectation value of f is:

$$\mathbb{E}_{x \sim P}[f(x)] = \int f(x) P(x) dx$$

Read the pieces: $f(x)$ is the thing you're measuring, $P(x)$ is how often you get each value, and the integral is an average of the first weighted by the second. The subscript names the distribution x is drawn from.

1.14.1 The mean

The simplest case is $f(x) = x$ — just measuring the variable itself:

$$\mathbb{E}[x] = \int x P(x) dx \equiv \mu$$

This is the **mean**, written with the Greek letter μ . For a discrete variable the integral is a sum over the possible values:

$$\mu = \sum_{x_i \in x} x_i P(x_i)$$

1.14.2 Variance and standard deviation

The mean tells you where the distribution sits. It doesn't tell you how tightly it's packed.

To measure spread, look at how far from the mean you typically land. You can't just average the distance, because the positive and negative deviations cancel. Square them first, and they don't:

$$V[x] = \mathbb{E}[(x - \mu)^2] = \mathbb{E}[x^2] - \mu^2 \equiv \sigma^2$$

The two forms are equal, and both get used. This is the **variance**, σ^2 . Its square root σ is the **standard deviation**.

Here is how to hold on to what σ means. The mean tells you the value to expect. The standard deviation tells you how close to that value your next measurement is likely to come.

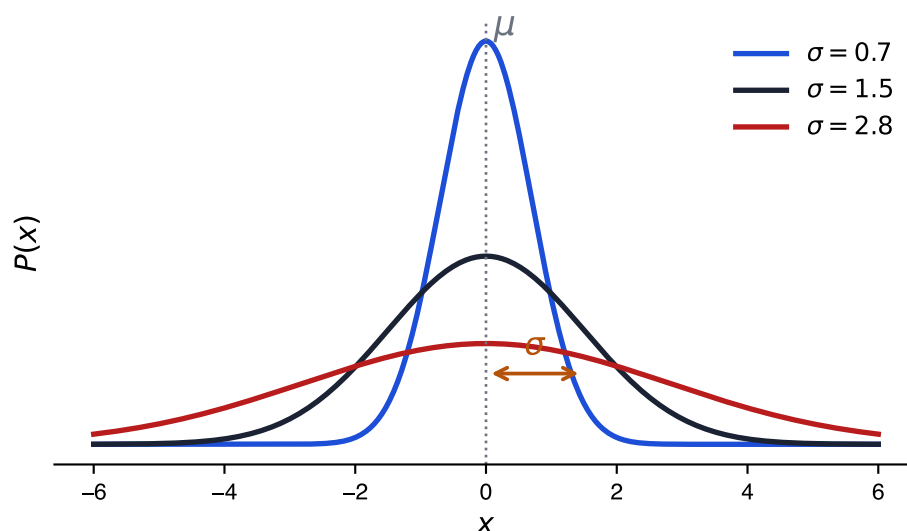


Figure 1.10: The same mean, three different spreads. μ says where the distribution is; σ says how far from it you should expect to land.

1.14.3 Covariance and correlation

For a two-dimensional distribution $P(x, y)$ there's an equivalent quantity capturing how the two move together:

$$\text{cov}[x, y] = \mathbb{E}[xy] - \mu_x \mu_y = \mathbb{E}[(x - \mu_x)(y - \mu_y)]$$

That second form says it plainly: if I move one feature away from its mean, how does the other one move?

Normalising by the two standard deviations gives the **correlation coefficient**:

$$\rho_{xy} = \frac{\text{cov}[x, y]}{\sigma_x \sigma_y}$$

which is dimensionless and easy to interpret. Suppose $\rho_{xy} = 0.5$. Take a data point sitting at the average of x , and move it up by one σ_x . The most likely thing is that y has moved up by half a σ_y . If ρ were 1, moving one by a standard deviation moves the other by a full standard deviation. If it's negative, moving one up moves the other down. Two highly correlated variables, in a scatter plot, lie close to a line.

The point of all these quantities is that they are single numbers you can compute for any distribution, without characterising the whole thing, and each of them tells you something real.

1.15 Distributions with names

Sometimes you don't have to measure a distribution at all, because you already know what it is. If you're flipping a coin, the distribution is known in advance — all you need is to measure its one parameter. When that's the situation, you can do much better than a histogram.

There are only a handful of these that come up constantly. Each answers a characteristic question.

1.15.1 Binomial — “what fraction?”

You run N independent trials, each of which succeeds or fails. Let n be the number of successes and p the underlying probability of success on any one trial. The probability of getting exactly n successes is:

$$f(n \mid N, p) = \frac{N!}{n! (N - n)!} p^n (1 - p)^{N - n}$$

with

$$\mathbb{E}[n] = Np \quad V[n] = Np(1 - p)$$

This is the distribution behind every “what fraction of...” question, which means it is behind an enormous amount of data science. True positive rate, false positive rate, efficiency, click-through rate, conversion rate — all binomial.

Usually you don't want n , you want the rate p , estimated as $\hat{p} = n/N$. Propagating the variance through that division:

$$V[\hat{p}] = V\left[\frac{n}{N}\right] = \frac{p(1 - p)}{N} \quad \Rightarrow \quad \sigma_{\hat{p}} = \frac{1}{\sqrt{N}} \sqrt{p(1 - p)}$$

That last formula is the error bar on every fraction you will ever quote. Learn it.

1.15.1.1 A worked example

Ask a class of 45 students how many watched the Super Bowl. Six hands go up.

Assume — and this assumption is doing real work — that these 45 are a random, representative sample of the university. Then the estimated rate is

$$\hat{p} = \frac{6}{45} = 0.13$$

and its uncertainty is

$$\sigma_{\hat{p}} = \frac{1}{\sqrt{45}} \sqrt{0.13 \times 0.87} = 0.05$$

So: 0.13 ± 0.05 . Thirteen percent, give or take five.

Now suppose someone at another university runs the same survey and reports 15%, and claims their students like football more. Do you believe them? No — not from this. Your own number has a five point uncertainty on it, and 15% is well inside that. The two results are perfectly consistent. Without the error bar you would have had a difference; with it you have nothing.

The interpretation of ± 0.05 is worth spelling out. If you repeated the whole exercise — pick 45 new students, count again — then about 68% of the time your answer would land within one σ of the truth, and about 95% of the time within two. Those numbers, 68 and 95, come from the normal distribution, which for large N is what the binomial becomes.

And you can run the formula backwards. If you need to know a rate to within half a percent, set $\sigma_{\hat{p}} = 0.005$, guess a rough value for p , and solve for N . That tells you how many people you need to survey — *before* you run the survey. This is how sample sizes get chosen.

1.15.2 Poisson — “how many?”

Take the binomial and push it to the limit of many trials each with tiny probability: $N \rightarrow \infty$, $p \rightarrow 0$, with the expected count $Np \rightarrow \nu$ staying finite. What comes out is the **Poisson distribution**:

$$f(n; \nu) = \frac{\nu^n}{n!} e^{-\nu}$$

with the remarkable property that its variance equals its mean:

$$\mathbb{E}[n] = \nu \quad V[n] = \nu \quad \Rightarrow \quad \sigma = \sqrt{\nu}$$

This is the distribution of counts and rates. Customers arriving at a store per hour, students walking through a door, decays of a radioactive sample, photons hitting a detector. Whenever you have counted n of something, your uncertainty on that count is:

$$n \pm \sqrt{n}$$

Look at what that implies. Count 100 things and your uncertainty is 10 — the absolute error grew. But as a *fraction* it went from nothing to 10%. Count 10,000 and the absolute error is 100 while the relative error is 1%. The relative uncertainty is

$$\frac{\sqrt{n}}{n} = \frac{1}{\sqrt{n}}$$

so precision improves as the square root of the sample size. More data always helps, and it always helps slowly. To halve your uncertainty you need four times the data.

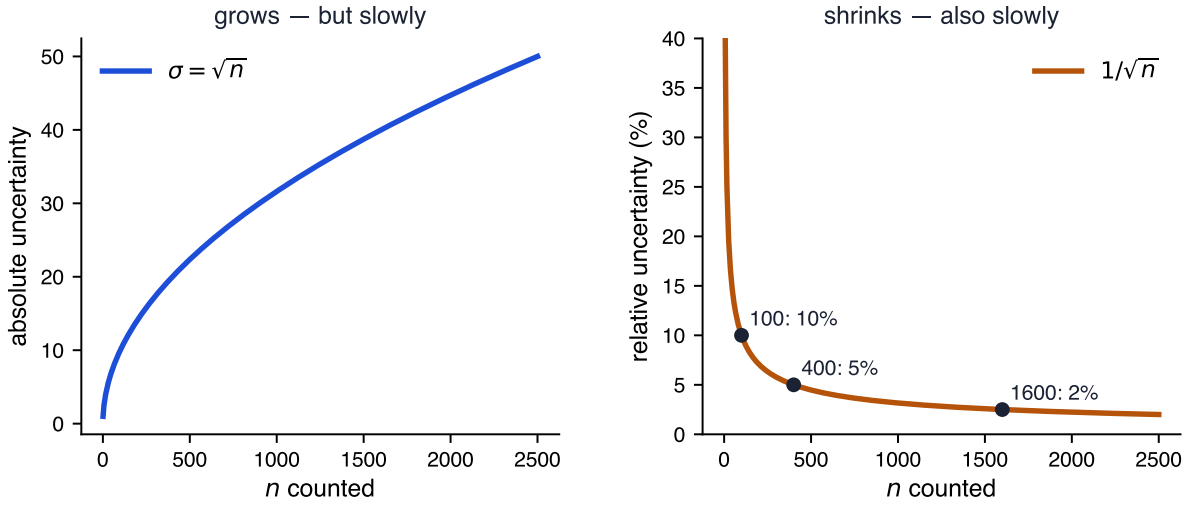


Figure 1.11: Counting statistics. The absolute uncertainty $\sqrt{n}$ grows with the count, but the relative uncertainty $1/\sqrt{n}$ shrinks — which is why more data helps, and why it helps slowly.

1.15.3 Exponential — “how long?”

Poisson tells you how many events fall in a window. The **exponential distribution** tells you how long you wait between them:

$$f(t; \tau) = \begin{cases} \frac{1}{\tau} e^{-t/\tau} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

with

$$\tau = \frac{1}{\nu} \quad \mathbb{E}[t] = \tau \quad V[t] = \tau^2 \quad \Rightarrow \quad \sigma = \tau$$

If customers arrive at a rate ν per hour, the time until the next one is exponentially distributed with mean $\tau = 1/\nu$. Note that the standard deviation equals the mean — the exponential is a very wide distribution, and “the average wait is ten minutes” is compatible with waiting a long time.

1.15.4 Normal — everything else

And then there’s the reason the other three are special cases.

The **central limit theorem** says that if you add or combine random variables — even ones drawn from different distributions — the result tends toward a **normal distribution**:

$$f(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi} \sigma} e^{-(x-\mu)^2/2\sigma^2}$$

with

$$\mathbb{E}[x] = \mu \quad V[x] = \sigma^2$$

Most quantities you measure are the combined result of many small contributions, which is why the normal distribution turns up everywhere. And it has a property that makes it especially convenient: it is completely specified by its mean and its standard deviation. If you know a distribution is normal, then μ and σ tell you *everything* about it. Nothing else is left to learn.

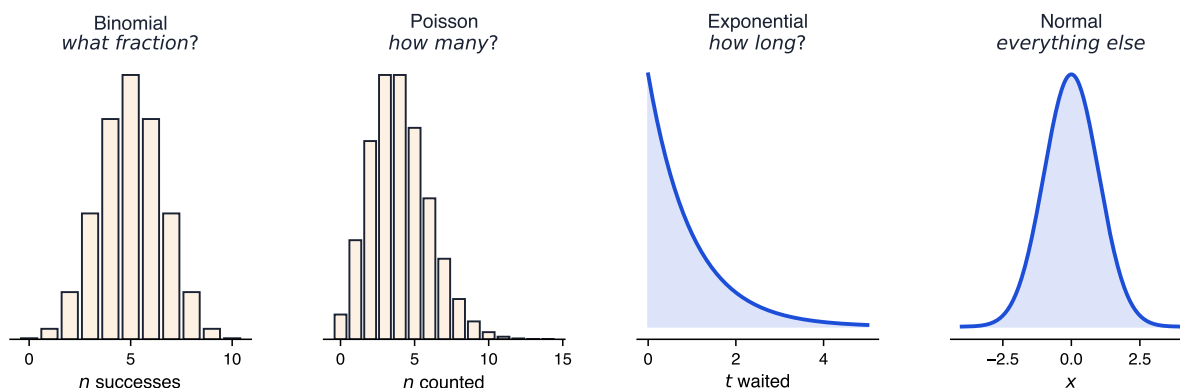


Figure 1.12: The four distributions worth knowing by name, and the question each one answers.

This is the payoff of the whole section. Identify which distribution you’re dealing with, measure its one or two parameters and their uncertainties, and you can compare, predict, and model. That’s the workflow.

1.15.5 What that buys you — curving an exam

Here is a use of the normal you have personally experienced.

Take an exam, and plot the distribution of scores across the class. If it’s roughly normal, it is described entirely by μ and σ . Now cut it at the mean and at one standard deviation either side. The properties of the normal distribution do the rest: 68% of the class lies within one σ of the mean, so 34% sits in each half of that band, and 16% lies beyond it on each side.

Label those four regions and you have a curve: 16% A, 34% B, 34% C, 16% D. Whether that’s a *good* way to assign grades is a separate argument. But it is a clean illustration of what the last paragraph claimed: two numbers, and every question about the distribution has an answer.

1.15.6 The caveat, which matters

Data often doesn’t look normal even when it “should,” and the reason is almost always the same one from the height–weight example: what you’re looking at is several normal distributions superimposed. Data science students and non-data science students. Smokers and non-smokers,

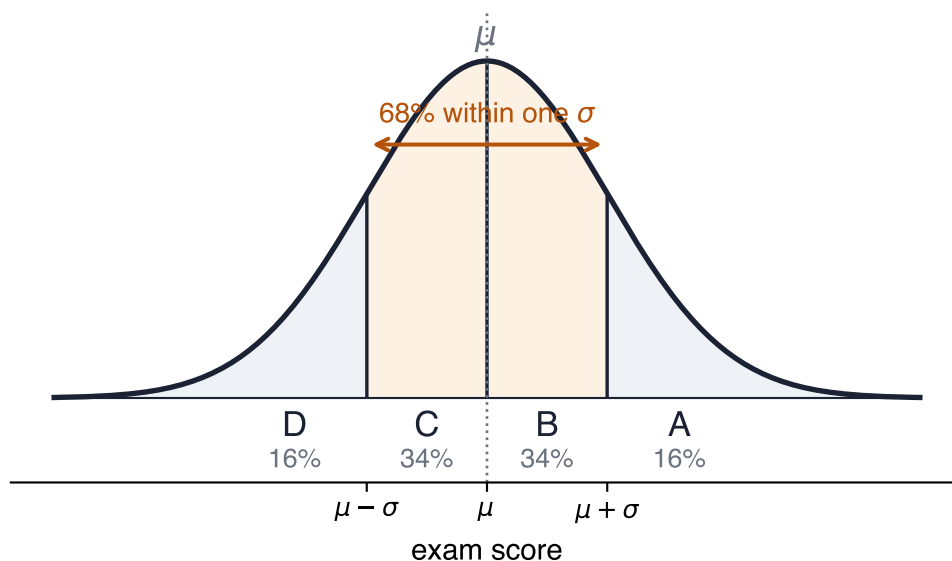


Figure 1.13: Curving to the distribution. The 68% within one standard deviation splits into 34% either side of the mean, leaving 16% in each tail.

whose insurance premiums form two clearly separated bumps. Each population is normal; the sum is not, and the mean of the sum is meaningless.

I see this constantly. Someone computes a mean and a standard deviation, reports them, and never plots the data — and if they had plotted it, they'd have seen immediately that the numbers don't describe it.

Always look at your data. When someone asks you for a mean and a standard deviation, ask yourself whether those two numbers are actually a fair characterisation of the distribution you have. There are formal tests for this. There's also looking at the histogram, which catches most of it and takes five seconds.

Including, it should be said, the exam scores from the previous section. Plot a class's marks and you will often find two bumps rather than one — the students who kept up and the students who didn't. Curving to a single μ and σ then describes neither group, which is a fair illustration of how quickly the caveat bites: the example was chosen precisely because it looks like a clean normal, and it frequently isn't.

1.16 Computing these things from data

Everything above is written in terms of $P(x)$ — but you don't have $P(x)$. You have a dataset. So how do you actually compute a mean?

You know the answer already: add up the numbers and divide by how many there are. What's worth seeing is *why* that is the same thing as $\int x P(x) dx$, because the connection is subtle and it is where the histogram formula earns its keep.

Keep two things distinct. The **set of possible values** $\mathcal{X} = \{x_1, x_2, \dots\}$ is what the sums in the probability formulas run over. Your **dataset** is a list of N observed values, in which the same value can appear many times.

Start from the definition and substitute the histogram estimate $P(x_i) \approx n(x_i)/(N\Delta x)$:

$$\mu = \int x P(x) dx \approx \sum_i x_i P(x_i) \Delta x = \sum_i x_i \frac{n(x_i)}{N\Delta x} \Delta x = \frac{1}{N} \sum_i x_i n(x_i)$$

Now read that last expression concretely. The sum runs over *possible values*. The value 1 appears 5 times, so it contributes 1×5 . The value 2 appears 3 times, contributing 2×3 . And so on — which is just adding up every number in your dataset, since a value appearing n times contributes itself n times. So:

$$\mu = \frac{1}{N} \sum_{x_i \in \text{dataset}} x_i$$

The familiar average. The underlying assumption that made it work is that **the histogram is a good estimator of the probability distribution** — which is the same assumption underneath everything else in this chapter.

The same substitution for the variance gives:

$$\sigma = \sqrt{\frac{1}{N} \sum_{x_i \in \text{dataset}} (x_i - \mu)^2}$$

1.16.1 Why $N - 1$

You will usually see that last formula with $N - 1$ in the denominator instead of N :

$$\sigma = \sqrt{\frac{1}{N - 1} \sum_i (x_i - \mu)^2}$$

This is the **Bessel correction**, and the reason for it is visible in the formula itself. To compute the standard deviation you need the mean — and you didn't know the mean either; you estimated it from the same data. That estimate consumed one piece of statistical information, one **degree of freedom**. Dividing by $N - 1$ rather than N accounts for it, and gives a better estimate of the true standard deviation.

For large N it makes almost no difference. For small N it does, which is exactly when you should be careful anyway.

1.17 A worked measurement

Let's put the whole chapter together on something small enough to see all at once.

Drop a ball from a height of one metre and measure how long it takes to hit the floor. From that, estimate g .

The physics is one line. Starting from rest, $y = y_0 + v_0 t + \frac{1}{2}at^2$ with $v_0 = 0$ and $a = -g$, so the distance fallen is

$$\Delta y = \frac{1}{2}g \Delta t^2 \quad \Rightarrow \quad g = \frac{2 \Delta y}{\Delta t^2}$$

With $\Delta y = 1$ m and $g = 9.8$ m/s², we'd expect $\Delta t = \sqrt{2\Delta y/g} \approx 0.45$ seconds.

Now go and do it. Not once — many times.

Trial	Δt (s)
1	0.47
2	0.42
3	0.49
4	0.58
...	...

The measurements are not the same. Your reaction time varies, the ball doesn't always leave your fingers the same way, the stopwatch has a resolution. Δt **is a random variable**, and that is the whole point of the exercise: a quantity you would have called a simple deterministic measurement turns out to have a distribution the moment you actually repeat it.

So histogram them. The centre of that histogram — the mean — is your estimate of Δt . Its width — the standard deviation — is your uncertainty. Push both through $g = 2\Delta y/\Delta t^2$ and you have not just a value for g but an error bar on it, and only then can you ask whether your measurement agrees with 9.8.

That is the entire scientific process in one page: estimate the parameter, quantify the uncertainty, test the agreement.

1.18 Generating data from a distribution

Everything so far has run in one direction: data comes in, and you work out what distribution it came from. Now run it backwards. **Given a distribution, produce data that follows it.**

That is the **Monte Carlo simulation** promised at the start of this chapter — the kind of generative model that knows the underlying probability distributions and *samples* them to make new data. It is how you test a method before you have real data, how you work out what a result would look like if a hypothesis were true, and, at the far end of this idea, roughly what a generative model is doing.

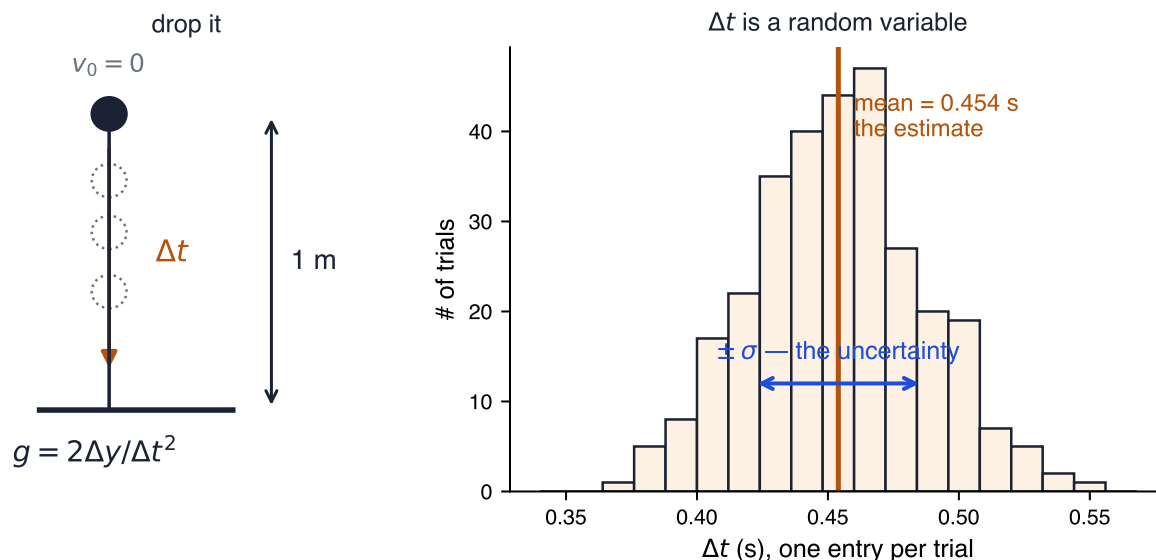


Figure 1.14: One metre, many drops. Each trial gives a slightly different Δt ; the histogram of trials has a centre, which is the estimate, and a width, which is the uncertainty.

The whole problem reduces to one question: **you have a source of uniform random numbers between 0 and 1. How do you turn those into draws from an arbitrary distribution?**

1.18.1 The uniform is the only thing you get

A computer gives you one distribution for free: the **uniform**, flat across its range,

$$P(x) = \frac{1}{\Delta x}$$

and it does not even really give you that. There are two ways to get random numbers out of a machine. You can harvest a genuinely random physical process — thermal noise, quantum measurements, the timing of your keystrokes — which your operating system does collect. But entropy arrives at whatever rate physics supplies it, so reading from such a source blocks and is rate-limited. Ask for ten million numbers for a simulation and you will wait.

So in practice you use a **pseudo-random** generator: a deterministic sequence that looks random. It starts from a **seed**, it is completely reproducible — same seed, same sequence, every time — and it eventually repeats. That determinism sounds like a defect and is usually a feature. A simulation run from a fixed seed does exactly the same thing next time, so a bug that shows up on the four-thousandth draw can be reproduced at all.

Everything below assumes you have that uniform source and asks how to bend it into the shape you want. There are two general answers.

1.18.2 Inversion

The first is **inverse transform sampling**, and it is a genuinely elegant trick.

Take the cumulative distribution of the density you want:

$$F(x) = \int_{-\infty}^x f(x') dx'$$

F climbs from 0 to 1, and provided it is continuous and increasing, it can be inverted. So: draw u uniformly between 0 and 1, and push it *backwards* through F .

$$u \sim \text{uniform}(0, 1) \quad x = F^{-1}(u) \quad \Rightarrow \quad x \sim f(x)$$

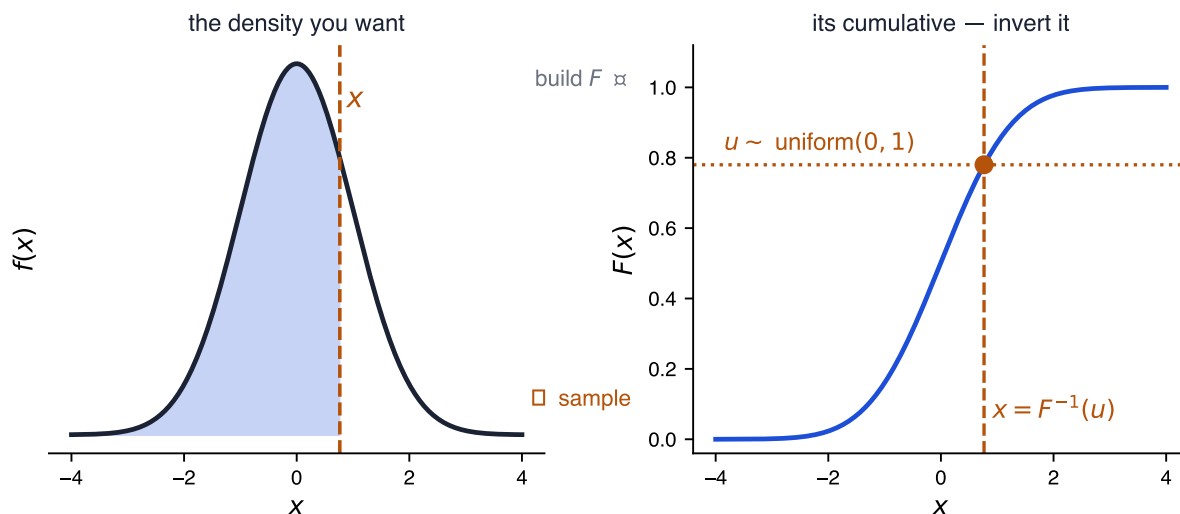


Figure 1.15: Inverse transform sampling. F maps each x to the probability lying to its left; choosing u uniformly picks a share of probability, and F^{-1} says which x owns it.

Why it works is worth seeing rather than accepting. $F(x)$ is the amount of probability lying to the left of x . Choosing u uniformly means choosing a *share of the probability* uniformly, and F^{-1} tells you which x that share belongs to. Where f is tall, F climbs steeply, so a wide band of u values maps into a narrow band of x — and that region catches more of your draws. Which is exactly what “distributed according to f ” means.

For the exponential the algebra comes out in three lines and gives $x = -\tau \ln(1-u)$; the notebook does it in full. When you can integrate and invert analytically, inversion is the best method there is: one uniform draw and one function evaluation per sample, and nothing wasted.

1.18.3 Accept/reject

The catch is that phrase — *when you can integrate and invert analytically*. Very often you can’t. You have a function you can **evaluate** at any point but cannot integrate in closed form,

which describes most functions and nearly every likelihood you will meet in practice. Inversion is unavailable, and you still want to sample.

Accept/reject needs no calculus at all. It needs only the ability to evaluate f , and it works like this:

Draw the box that contains the function — x from x_{min} to x_{max} , y from 0 up to some y_{max} above the curve's peak. Now throw darts uniformly into that box. For each dart at (x, y) , ask one question: **did it land under the curve?** That is, is $y < f(x)$? If it did, keep its x . If not, throw it away and try again. Stop when you have as many kept values as you wanted.

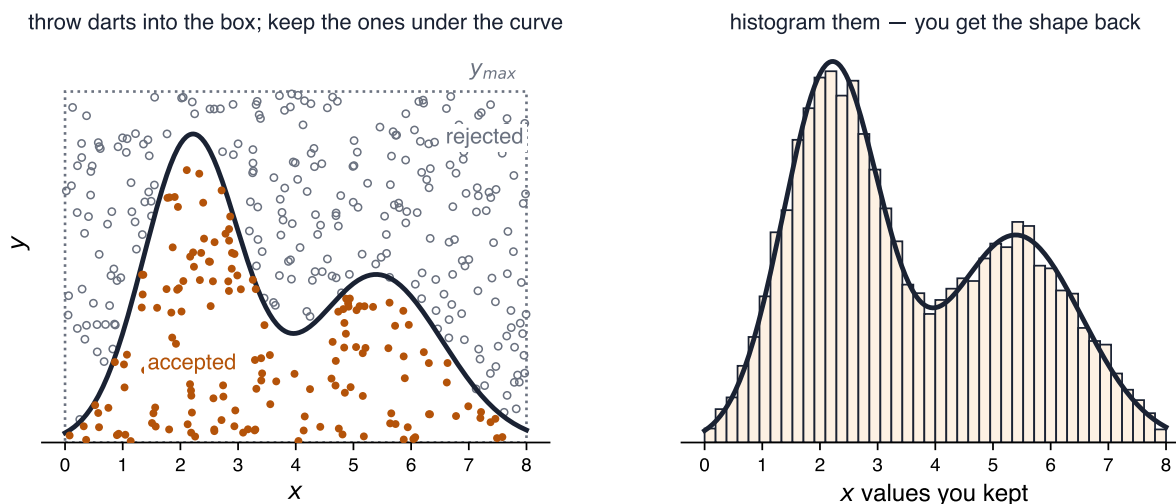


Figure 1.16: Accept/reject. Darts land uniformly in the box; the ones under the curve are kept. Because a tall region of f covers more of its vertical strip, it keeps proportionally more darts — so the x values you keep are distributed like f .

The kept x values are distributed like f , and the picture shows why. Look at a thin vertical strip at some x . The darts in that strip are spread uniformly up the box, so the fraction surviving is the fraction of the strip lying under the curve — which is $f(x)/y_{max}$, proportional to the height of the curve there. Tall regions keep more darts, low regions keep fewer, in exact proportion to f . That is the definition of sampling from f .

The right-hand panel is the check, and it is the one worth building the habit of: histogram the values you kept and the shape of f comes back. A sampler that is subtly wrong — a range mis-specified, a comparison the wrong way round — still returns a perfectly plausible list of numbers. The histogram is what gives it away.

The cost is the rejected darts. Every one is a wasted evaluation. If f is a narrow spike in a wide box, most of the box is empty, and you might throw away 99 darts for every one you keep. That is the trade: inversion is efficient but needs an integral you can do; accept/reject asks almost nothing of the function but pays for it in wasted draws. Choosing between them is a real decision, and it comes up whenever you simulate anything.

1.18.4 And integration, for free

Now the part that makes accept/reject more than a sampling trick.

You just threw darts uniformly into a box whose area you know exactly. The fraction landing under the curve is the fraction of the box's area that lies under the curve. So the area under f — which is to say **the integral of f** — is:

$$\int_{x_{\min}}^{x_{\max}} f(x) dx = A \times \frac{\# \text{ accepts}}{\# \text{ trials}} \quad \text{where} \quad A = (x_{\max} - x_{\min})(y_{\max} - y_{\min})$$

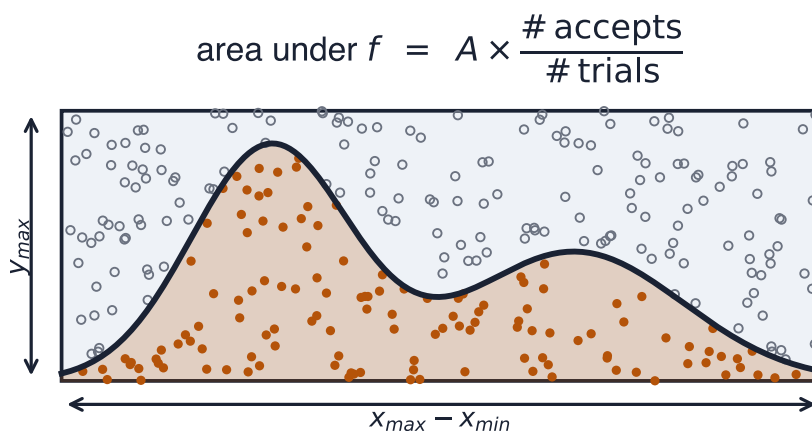


Figure 1.17: Monte Carlo integration. The same darts, counted rather than kept: the fraction landing under the curve is the fraction of the box's area that lies beneath it.

This is **Monte Carlo integration**, and it is one of the most useful techniques in computational science. It requires no calculus. It works on functions you can only evaluate. And unlike grid-based numerical integration, its cost does not explode as you add dimensions — which is why it is the standard tool for high-dimensional integrals, and therefore for most of the integrals that actually arise.

Its accuracy is something you can already work out, because it is counting statistics from earlier in this chapter. You accepted n darts, so the uncertainty on that count is $\sqrt{n}$, and the relative uncertainty on the integral is $1/\sqrt{n}$. Want one more decimal place? Throw a hundred times as many darts. The same square-root law that governed the Super Bowl survey governs this.

Which is a fitting place to end. The chapter opened with a rain forecast and a question about what a probability means. It closes with the observation that if you can evaluate a function and generate uniform random numbers, you can both sample it and integrate it — and that you know, in advance, exactly how wrong the answer will be.

1.19 What's next

You now have the language. You know what a probability is, both readings of it, how to update one with Bayes' theorem, what a random variable and a distribution are, how a histogram estimates a distribution, how to summarise one with a mean and a variance, which named distributions to reach for and what each one's error bar looks like.

What you can't yet do is run any of it.

The next chapter is a notebook, and it turns the last section into working code: a pseudo-random number generator built from scratch, the exponential's inversion carried through in full, a histogram function written by hand so you know exactly what `np.histogram` is doing, and accept/reject and Monte Carlo integration implemented as the same loop.

The reasoning is done. What remains is to make it run — and to find out how much of it you believed only because it was written down.

